

Parametric Equations

ANSWER KEY



Eliminating the parameter

- 1 Consider the curve $x = 2t + 1$, $y = t^2$.
 - a Eliminate the parameter to write y as a function of x . $t = \frac{x-1}{2}$, so $y = \frac{(x-1)^2}{4}$.
 - b Identify the curve. A parabola with vertex $(1, 0)$ opening upward.
 - 2 Eliminate the parameter for $x = 3\cos t$, $y = 3\sin t$, $0 \leq t < 2\pi$, and describe the curve.
 $x^2 + y^2 = 9\cos^2 t + 9\sin^2 t = 9$: a circle of radius 3 centered at the origin, traced counterclockwise.
 - 3 For the line $x = t - 2$, $y = 2t + 3$:
 - a Find the points corresponding to $t = 0$ and $t = 2$. $t = 0$: $(-2, 3)$; $t = 2$: $(0, 7)$.
 - b Eliminate the parameter to write the Cartesian equation. $t = x + 2$, so $y = 2x + 7$.
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Projectile motion

- 4 A projectile follows $x = 40t$, $y = -5t^2 + 30t$ (metres, seconds).
 - a When does the projectile land ($y = 0$, $t > 0$)? $-5t(t - 6) = 0$, so $t = 6$ s.
 - b Find the horizontal range. $x(6) = 240$ m.
 - c Find the maximum height. Peak at $t = 3$: $y(3) = -45 + 90 = 45$ m.
 - 5 A second projectile is launched with $x = 25t$, $y = -5t^2 + 40t$. Find its time of flight and maximum height.
Land: $-5t(t - 8) = 0$, so $t = 8$ s. Peak at $t = 4$: $y(4) = -80 + 160 = 80$ m.
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Ellipses and other parametric curves

- 6 Eliminate the parameter for $x = 4\cos t$, $y = 2\sin t$, and identify the resulting curve.
 $\frac{x}{4} = \cos t$, $\frac{y}{2} = \sin t$: $\frac{x^2}{16} + \frac{y^2}{4} = 1$ — an ellipse with semi-axes 4 and 2.
 - 7 Eliminate the parameter for $x = t^2 - 1$, $y = t^3$ by expressing t in terms of x where possible, or by finding a relation between x and y directly.
From $x = t^2 - 1$: $t^2 = x + 1$. Squaring $y = t^3$: $y^2 = t^6 = (t^2)^3 = (x + 1)^3$. Relation: $y^2 = (x + 1)^3$.
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Finding the parametric derivative

- 8 For the curve $x = t^2$, $y = 4t$, find $\frac{dy}{dx}$ in terms of t using $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$.
 $\frac{dx}{dt} = 2t$, $\frac{dy}{dt} = 4$: $\frac{dy}{dx} = \frac{4}{2t} = \frac{2}{t}$ (for $t \neq 0$).
- 9 For the curve $x = \cos t$, $y = \sin t$, find the slope of the tangent line at $t = \frac{\pi}{4}$.
 $\frac{dx}{dt} = -\sin t$, $\frac{dy}{dt} = \cos t$: $\frac{dy}{dx} = -\cot t$. At $t = \frac{\pi}{4}$: slope = $-\cot \frac{\pi}{4} = -1$.

More parametric practice

- 10 A Ferris wheel of radius 10 m has its center 12 m above the ground and completes one revolution every 40 seconds. Write parametric equations for the height y of a rider above the ground as a function of time t (seconds), assuming the rider starts at the bottom of the wheel.
 Angular speed $\omega = \frac{2\pi}{40} = \frac{\pi}{20}$. Starting at the bottom means the vertical position begins at a minimum, matching $-\cos$: $y(t) = 12 - 10\cos\left(\frac{\pi}{20}t\right)$.
- 11 Two particles travel along paths $x_1 = t$, $y_1 = t^2$ and $x_2 = 4 - t$, $y_2 = 3t - 2$, sharing the same parameter t (time). Find the value of t at which they collide, and the point of collision.
 Need $x_1 = x_2$: $t = 4 - t$, so $t = 2$. Check $y_1 = y_2$ at $t = 2$: $y_1 = 2^2 = 4$; $y_2 = 3(2) - 2 = 4$. Both match, confirming a genuine collision at $t = 2$, at the point $(2, 4)$.

Graphing a parametric curve after elimination

- 12 For $x = t$, $y = t^2 - 2$ (with t ranging over all reals), the eliminated Cartesian form $y = x^2 - 2$ is shown. Use the graph to describe the direction the curve is traced as t increases.
 As t increases from $-\infty$ to ∞ , the curve is traced from the upper-left, down through the vertex $(0, -2)$, then up to the upper-right — left to right along the parabola.

Restricting the parameter domain

- 13 The curve $x = \cos t$, $y = \sin t$ for $t \in [0, \pi]$ traces only part of the unit circle. Describe which part, and state the domain and range restrictions this implies on x and y .
 For $t \in [0, \pi]$, $y = \sin t \geq 0$, so only the UPPER half of the unit circle is traced (from $(1, 0)$ counterclockwise to $(-1, 0)$). This restricts $y \geq 0$ while x still ranges over $[-1, 1]$.

Parametrizing a line segment

- 14 Write parametric equations for the line segment from $A(2, -1)$ to $B(6, 5)$, using $t \in [0, 1]$ (so $t = 0$ gives A and $t = 1$ gives B).
 $x(t) = 2 + 4t$, $y(t) = -1 + 6t$ for $t \in [0, 1]$ (using $x = x_1 + t(x_2 - x_1)$, $y = y_1 + t(y_2 - y_1)$).

- 15 Using the parametrization above, find the point at $t = 0.25$ (one-quarter of the way from A to B).
 $x(0.25) = 2 + 1 = 3$; $y(0.25) = -1 + 1.5 = 0.5$. Point: $(3, 0.5)$.
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Parametric speed and motion review

- 16 A particle's position is $x(t) = 3t$, $y(t) = 4t$ for $t \geq 0$ (measured in meters, seconds). Find the particle's speed, and confirm it matches $\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}$.
 $\frac{dx}{dt} = 3$, $\frac{dy}{dt} = 4$. Speed $= \sqrt{9 + 16} = 5$ m/s (constant, matching straight-line motion at constant velocity).